MedLink: Smart Pre-Visit Triage, Booking & Patient File

Graduation Project Proposal · October 2026

1. Summary

MedLink is a medical web app that prepares the visit instead of only booking it. Patients report their symptoms and receive an initial triage of urgency and a suggested medical specialty, then compare doctors by specialty and consultation fee, book an appointment, and pay online or use their health insurance, depending on coverage and accepted insurance providers. The doctor writes and maintains the patient's medical history, and receives an AI-generated draft summary and possible differential diagnoses to consider.

> **Core principle:** AI never makes the final diagnosis and never replaces the doctor. Triage is the core of MedLink: before any booking, fixed rules reviewed by physicians check red flags and assign an acuity level (Emergency, Urgent or Routine), and the system suggests the appropriate specialty with its clinical rationale, so every patient is directed to the right care at the right time. Triage is initial guidance, not a final diagnosis.

Key components

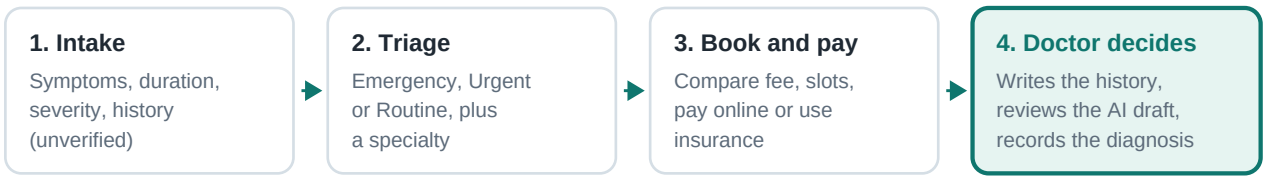
Component	What it provides
Smart triage	Three acuity levels, adaptive questions, risk modifiers (age, pregnancy, comorbidities) and a specialty with its rationale (section 3.1)
Booking and payment	Compare doctors by fee, slots and accepted insurance, then pay online or use health insurance (section 3.2)
Doctor-written patient history	One continuous record written by the doctor; what the patient enters stays unverified until the doctor adds it (section 3.3)
Doctor's final decision	After the doctor takes the history, the AI prepares a draft summary and suggests differential diagnoses; the doctor reviews the AI-generated draft, performs a clinical examination, proposes an action plan (medical and paramedical), and then records the final diagnosis (section 3.4)

2. The Problem

- Patients do not know which specialty to see or how urgent their case is, and cannot compare doctors' prices or insurance.
- Doctors get a few minutes per patient and an unstructured, spoken history.
- Existing platforms only find a doctor and a time slot; they do not prepare the visit.

3. The Four Pillars

The doctor stays in charge at every step of the care flow



Patient input stays unverified until the doctor adds it to the history.

The flow runs left to right; the doctor, not the AI, owns the history and the final diagnosis.

3.1 Smart Triage

The patient enters presenting symptoms, onset and duration, severity and past medical history, with **adaptive follow-up questions**. Fixed rules reviewed by physicians check red flags and assign a preliminary triage acuity level: **Emergency** (booking is paused and the patient is referred to the emergency department), **Urgent** (clinical review within 24 to 48 hours) or **Routine**. Age, pregnancy, comorbidities and uncertain answers can raise the acuity level. The system then suggests the **appropriate specialty with its clinical rationale**. Triage is initial guidance, not a final diagnosis.

3.2 Doctor Selection, Booking and Payment

The patient compares doctors by specialty, available slots, **consultation fee** and **accepted insurance**, then selects an appointment slot and **pays online or uses health insurance** according to the coverage and eligibility verification steps; the co-payment is paid at the clinic.

3.3 Medical History and Patient Draft

Each patient has one continuous medical file whose **history is written by the doctor**: past medical history, clinical notes and recorded diagnoses from previous encounters. What the patient enters (presenting symptoms, duration, severity and self-reported medical history) is **unverified input for triage** until the doctor adds it to the history. Before the clinical examination, AI prepares a **draft clinical summary** from current and previous information; the doctor reviews and edits it and writes the **final diagnosis**. Every addition or change to the history is recorded in the audit log.

3.4 Doctor's Final Decision

The doctor reviews the patient record, the triage result and the AI-generated draft, and may consider AI-suggested **differential diagnoses** (ranked, each with its supporting clinical findings and missing information, which the doctor may accept, reject or disregard). The doctor documents the clinical notes, proposes an action plan (medical, such as the prescription and referrals, and paramedical, such as physiotherapy, nursing care or laboratory and imaging requests), and **records the final diagnosis**.

4. Role of Artificial Intelligence

The AI does	The AI does not
Summarize and organize the history into a draft	Make the final diagnosis
Highlight missing information	Prescribe medication or doses, or make treatment decisions
Suggest ranked differential diagnoses for the doctor to review	Replace the doctor's judgment or confirm a diagnosis on its own

Safety rules

- Triage and emergency warnings come from fixed, physician-reviewed rules that run before and independently of any AI.
- Every AI output is a labeled, editable draft that uses only entered data, and each point traces back to a patient answer or history item.
- Differential diagnoses are a ranked list, never one confident answer, and the physician advisors review their content and ranking rules.
- Specialty suggestion starts as a transparent, physician-reviewed scoring table; machine learning can be added later and validated against doctors' judgment.

5. Main Features

Patient: account, adaptive intake, triage result with emergency alerts, specialty suggestion, doctor comparison and booking, online or insurance payment with receipt, and a view of the confirmed history and diagnosis.

Doctor: profile, availability, fee and accepted insurers; patient file whose history the doctor writes, with past visits; AI draft summary; differential diagnoses; notes, action plan and final diagnosis.

Hospital / clinic: hospital profile with its departments and clinics, doctors' schedules, fees and accepted insurers per department, and an overview of booked appointments; triage and the specialty suggestion direct each patient to the right department, which reduces visits to the wrong clinic and keeps appointments organized.

Medication reminders: the doctor enters a structured medication schedule (drug, dose, frequency, start date and duration), and the system sends the patient a reminder at each scheduled time through web push, with email as a backup, with "taken" and "snooze" actions. Reminders only relay what the doctor prescribed: the AI never suggests medication or doses, and the patient cannot change them. The patient opts in, and the notification text stays generic so drug details are not shown on the lock screen.

Medication list and prescription: the doctor writes the prescription by choosing each drug from a built-in medication list (trade name, active ingredient, form and strength) with autocomplete, or by typing a drug that is not listed, then sets the dose, frequency, start date, duration and instructions. The list is prepared for the project and reviewed with the physician advisors. The system turns each prescription item into scheduled dose reminders; it does not check or suggest doses, so clinical responsibility stays with the doctor, and every change is recorded in the audit log.

Platform: Arabic and English interface with right-to-left support, role-based access, audit log, encryption in transit and at rest, and a third-party payment gateway so card details are never stored.

6. Conclusion

MedLink makes the visit better prepared through a safer triage, a transparent choice of doctor and payment, and a patient history the doctor can trust and keep up to date, with AI strictly in a supporting role and the doctor making every clinical decision.